Computational efficiency of multiresolution topology optimization for the MBB beam

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Abstract

This report investigates the computational efficiency of the multiresolution topology optimization (MTOP) approach for a two-dimensional minimum-compliance MBB beam. The MTOP formulation uses a coarse finite element analysis mesh and a finer density mesh. Its convergence behavior, computation time, and optimized layouts are compared with a classical density-based topology optimization formulation at the same final density resolution.

1 Problem statement

The aim of this assignment is to assess whether the MTOP approach introduced by Nguyen et al. [1] can reduce computation time for two-dimensional minimum-compliance topology optimization without significantly changing the optimized structural layout.

The benchmark problem is the half MBB beam from the lecture code. The classical formulation is solved on a fine finite element mesh, while the MTOP formulation performs finite element analysis on a coarser mesh and represents the material distribution on a finer density mesh.

2 Structural model

2.1 Geometry and discretization

The structure is modeled as a two-dimensional half MBB beam. The classical baseline uses a finite element mesh of 600×200 bilinear quadrilateral elements. The MTOP model uses a finite element mesh of 120×40 elements and a density mesh of 5×5 density cells per finite element, resulting in the same 600×200 density resolution as the classical model.

Table 1: Main discretization parameters.

Model	FE mesh	Density cells per FE	Density mesh
Classical	600×200	1×1	600×200
MTOP	120×40	5×5	600×200

2.2 Boundary conditions and loading

The boundary conditions and load are taken from the half MBB beam lecture implementation. The beam is loaded by a vertical point force at the upper-left node of the half model. Symmetry and support constraints are imposed as in the Chapter 9 lecture code.

2.3 Material interpolation

The density-based topology optimization uses SIMP interpolation:

$$E(\rho_e) = E_{\min} + \rho_e^p (E_0 - E_{\min}), \tag{1}$$

where ρ_e is the physical element density, $p = 3$ is the penalization power, $E_0 = 1$, and $E_{\min} = 10^{-9}$.

3 Optimization problem

The minimum-compliance topology optimization problem is formulated as

$$\begin{aligned}
\min_{\mathbf{x}} \quad & C(\mathbf{x}) = \mathbf{F}^T \mathbf{U} && (2)\\
\text{subject to} \quad & \mathbf{K}(\mathbf{x})\mathbf{U} = \mathbf{F}, && (3)\\
& \frac{1}{N}\sum_{i=1}^{N} \rho_i \leq V^*, && (4)\\
& 0 \leq x_i \leq 1, && (5)
\end{aligned}$$

where $\mathbf{x}$ contains the design variables, ρ_i are the physical densities, $V^* = 0.5$ is the prescribed volume fraction, and N is the number of density cells or finite elements depending on the formulation.

4 Filtering and projection

4.1 Sensitivity filtering

The first study uses sensitivity filtering with a filter radius of 24 elements on the 600×200 density scale. The filtered sensitivity is computed from neighboring density cells within the filter radius.

4.2 Density filtering

The density-filtered formulation first maps design variables to filtered densities and then evaluates the objective, constraint, and sensitivities with respect to the original design variables through the filter operation.

4.3 Heaviside projection

For the MMA experiments, Heaviside projection is applied after density filtering:

$$\bar{\rho}_i = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\rho}_i - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \tag{6}$$

where $\tilde{\rho}_i$ is the filtered density, η is the projection threshold, and β controls projection sharpness.

5 Solution strategy

5.1 Classical OC baseline

The first benchmark solves the classical MBB problem using the optimality criteria method with:

- mesh size 600×200,
- volume fraction 0.5,
- penalization power 3,
- sensitivity filter radius 24.

The convergence history is recorded both by iteration number and by elapsed wall-clock computation time. The implementation follows the lecture script structure: the physical density is equal to the design density for sensitivity filtering, the compliance sensitivities are filtered, and the OC bisection enforces the volume constraint at each iteration.

5.2 MTOP with optimality criteria

The MTOP implementation keeps a fine density representation while solving equilibrium on a coarser analysis mesh. Each 120×40 finite element contains 5×5 density cells. The stiffness contribution of each analysis element is computed from the density cells assigned to that element.

For an analysis element e, let $\mathcal{D}_e$ be the set of its 25 density cells. The SIMP scaling of the element stiffness is computed as the average of the penalized density values,

$$\alpha_e = \frac{1}{25} \sum_{i \in \mathcal{D}_e} x_i^p, \tag{7}$$

so that

$$\mathbf{K}_e = (E_{\min} + (E_0 - E_{\min})\alpha_e)\,\mathbf{K}_0. \tag{8}$$

The compliance sensitivity for a density cell i inside analysis element e is therefore

$$\frac{\partial C}{\partial x_i} = -\frac{p}{25}(E_0 - E_{\min}) x_i^{p-1} \mathbf{u}_e^T \mathbf{K}_0 \mathbf{u}_e. \tag{9}$$

These sensitivities are then filtered on the full 600×200 density grid using the same sensitivity filter radius as in the classical baseline.

5.3 MTOP with MMA

The final study replaces the optimality criteria update by MMA. The MMA experiments are performed for sensitivity filtering, density filtering, and Heaviside projection with several projection thresholds.

6 Sensitivity verification

The implementation will be verified by comparing analytical sensitivities with finite-difference approximations for representative design variables. Both compliance and volume sensitivities will be checked before the final comparison between the classical and MTOP formulations is made.

7 Numerical experiments

7.1 Experiment matrix

7.2 Convergence criteria and timing

For each experiment, the following quantities are stored per iteration:

Table 2: Planned numerical experiments.

Case	Approach	Optimizer	Filter or projection
1	Classical	OC	Sensitivity filter
2	MTOP	OC	Sensitivity filter
3	Classical	OC	Density filter
4	MTOP	OC	Density filter
5	MTOP	MMA	Sensitivity filter
6	MTOP	MMA	Density filter
7	MTOP	MMA	Heaviside projection, varying η

- compliance,
- volume fraction,
- maximum design change,
- elapsed wall-clock time,
- optimizer status or continuation parameters.

All timing comparisons should be made on the same machine and with the same MATLAB version.

8 Results

8.1 Classical OC baseline

The classical OC baseline for assignment step 1 converged after 94 iterations. The final compliance was 224.5961, the final volume fraction was 0.500074, and the final maximum design change was 9.95×10^{-3}. The measured wall-clock time on the machine used for this run was 241.99 s.

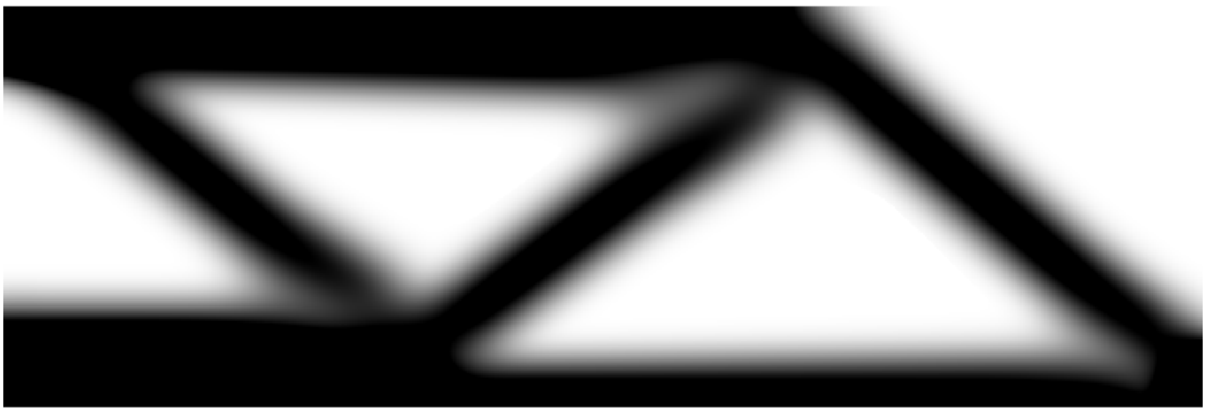

Figure 1: Optimized classical OC design with sensitivity filtering for a 600×200 mesh, volume fraction 0.5, penalization power 3, and filter radius 24 elements. Black denotes material and white denotes void.

8.2 MTOP OC with sensitivity filtering

The MTOP OC run converged after 93 iterations. The final MTOP compliance, evaluated on the coarse analysis mesh, was 219.0052. To compare the design on the same analysis model as

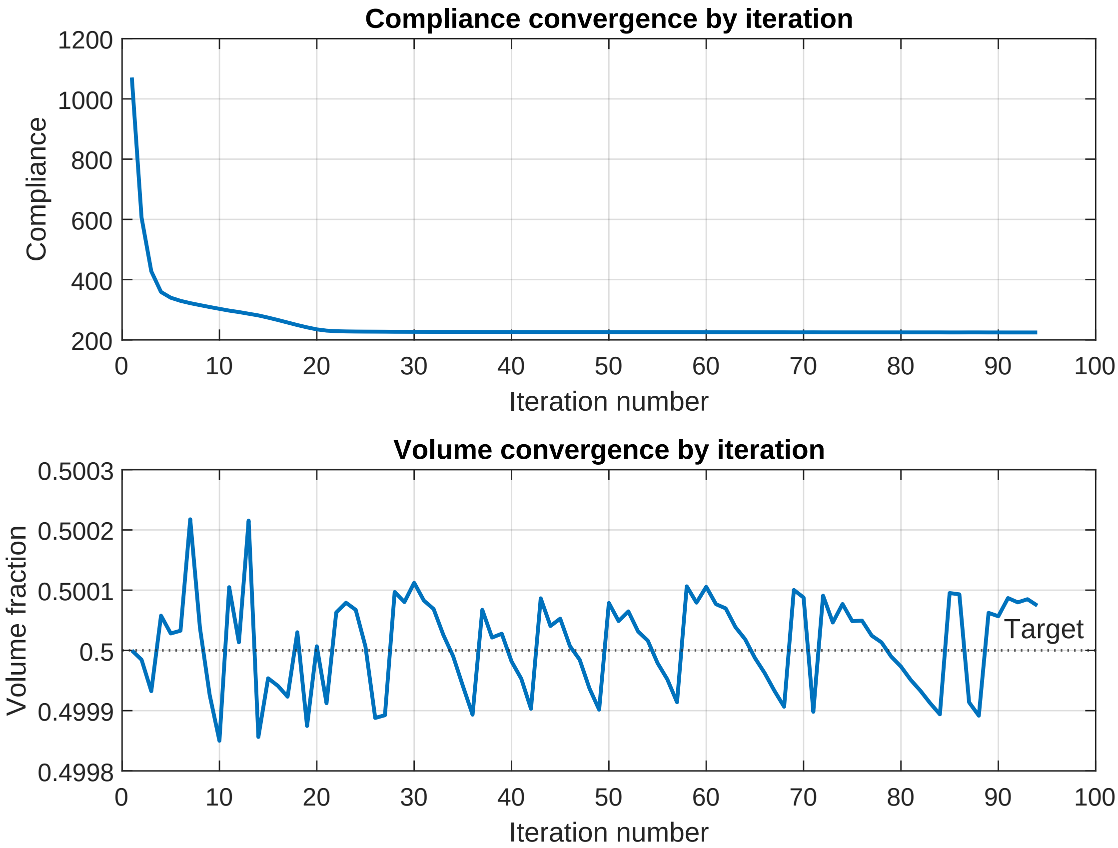

Figure 2: Classical OC convergence history as a function of iteration number. The compliance decreases rapidly during the first iterations and then approaches a plateau near the final design. The volume fraction remains close to the prescribed value throughout the OC updates.

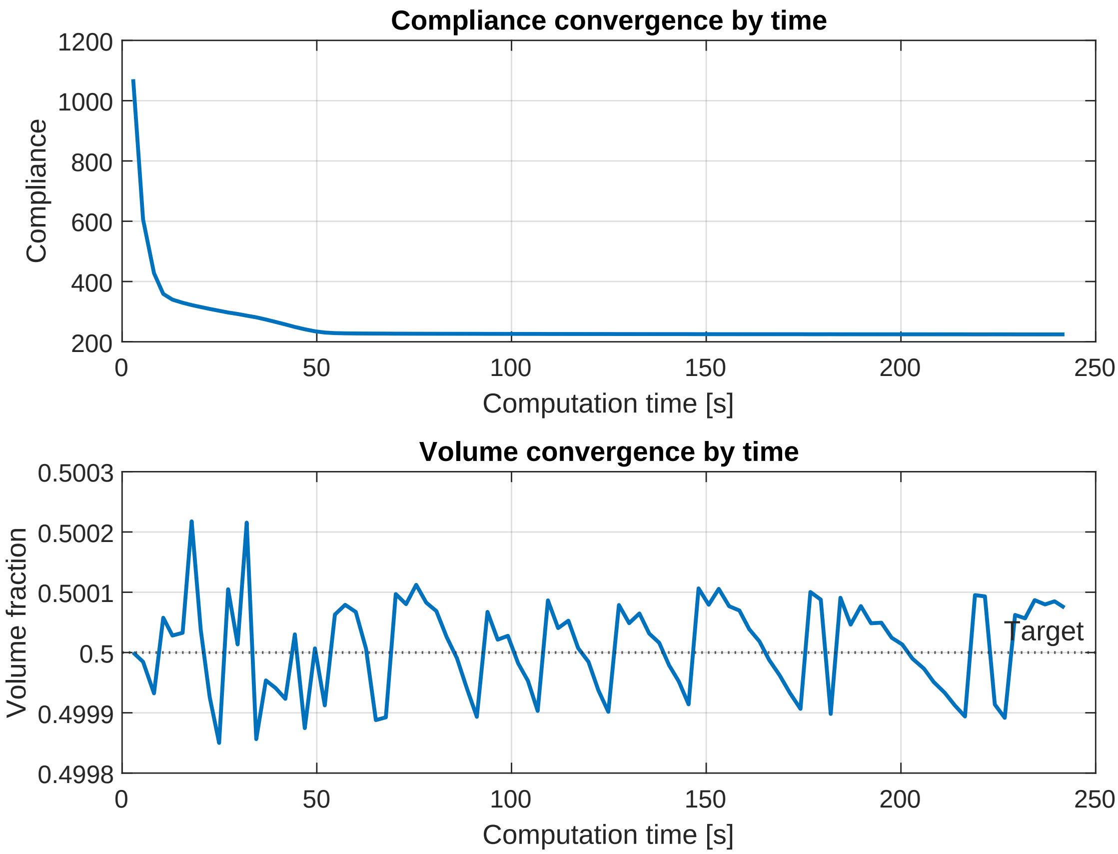

Figure 3: Classical OC convergence history as a function of elapsed computation time. The same compliance and volume histories are shown against wall-clock time to support the later runtime comparison with the MTOP approach.

the classical baseline, the final MTOP density field was also evaluated on the 600×200 finite element mesh. This gave a fine-mesh compliance of 224.6763, only 0.0357% higher than the classical value of 224.5961. The final volume fraction was 0.500017, and the measured MTOP optimization time was 8.82 s.

Figure 4: Optimized MTOP OC design with a 120×40 finite element mesh and 5×5 density cells per finite element. Black denotes material and white denotes void.

8.3 Density filtering comparison

Results for this assignment step will be added after the density-filtered classical and MTOP runs are completed.

8.4 MMA and Heaviside projection

Results for this assignment step will be added after the MMA and Heaviside projection studies are completed.

8.5 Runtime comparison

Table 3: Summary of runtime and final objective values.

Case	Iter.	Time [s]	Native C	Fine C	Volume
Classical OC sensitivity	94	241.99	224.596	224.596	0.500074
MTOP OC sensitivity	93	8.82	219.005	224.676	0.500017
Classical OC density	–	–	–	–	–
MTOP OC density	–	–	–	–	–
MTOP MMA sensitivity	–	–	–	–	–
MTOP MMA density	–	–	–	–	–
MTOP MMA Heaviside	–	–	–	–	–

9 Discussion

The step 1 design shows the expected half MBB beam load paths: material is concentrated in the main chord regions and in diagonal members that transfer the applied load toward the support. The relatively large sensitivity filter radius produces smooth, wide members and suppresses checkerboard patterns. The volume history confirms that the OC update keeps the design close

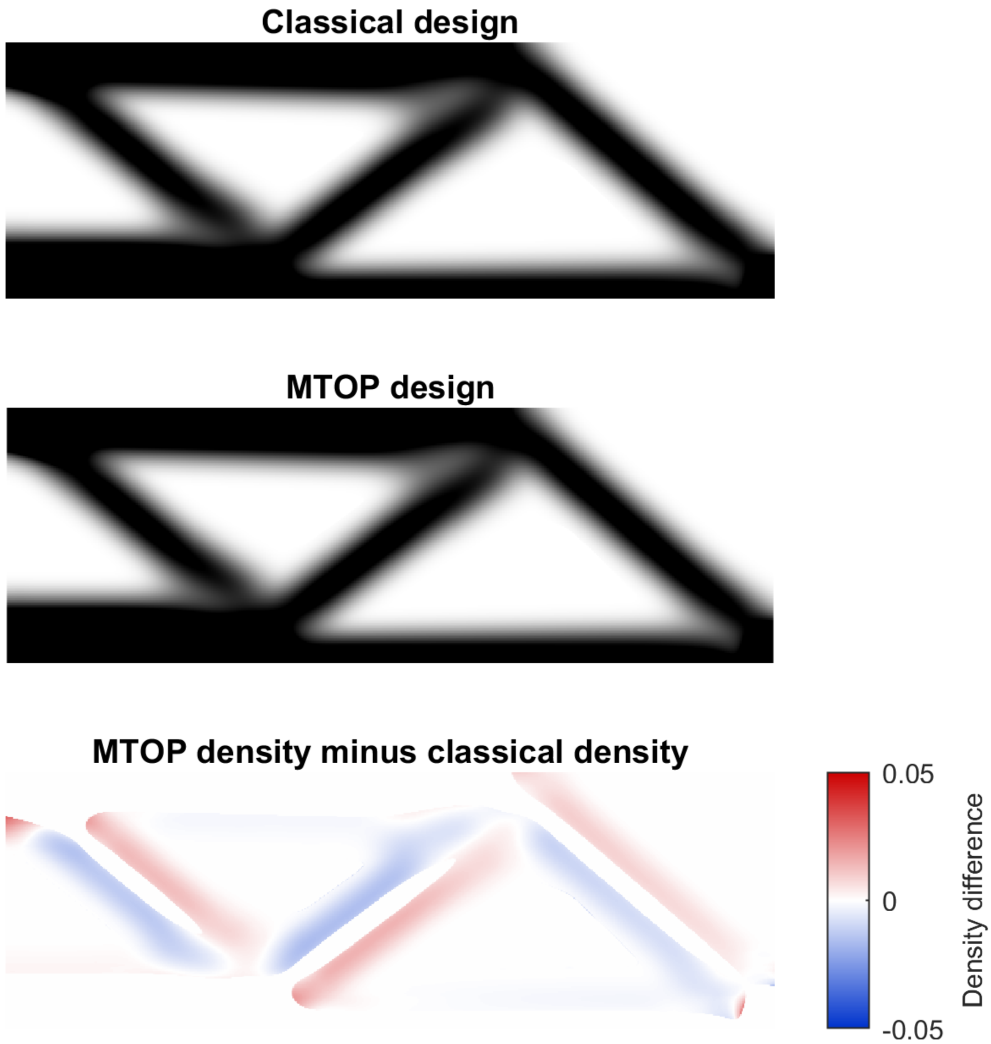

Figure 5: Comparison between the classical and MTOP OC designs. The signed difference plot shows $\rho_{\mathrm{MTOP}} - \rho_{\mathrm{classical}}$ on a narrowed color scale, because the maximum absolute density difference is only 0.03035.

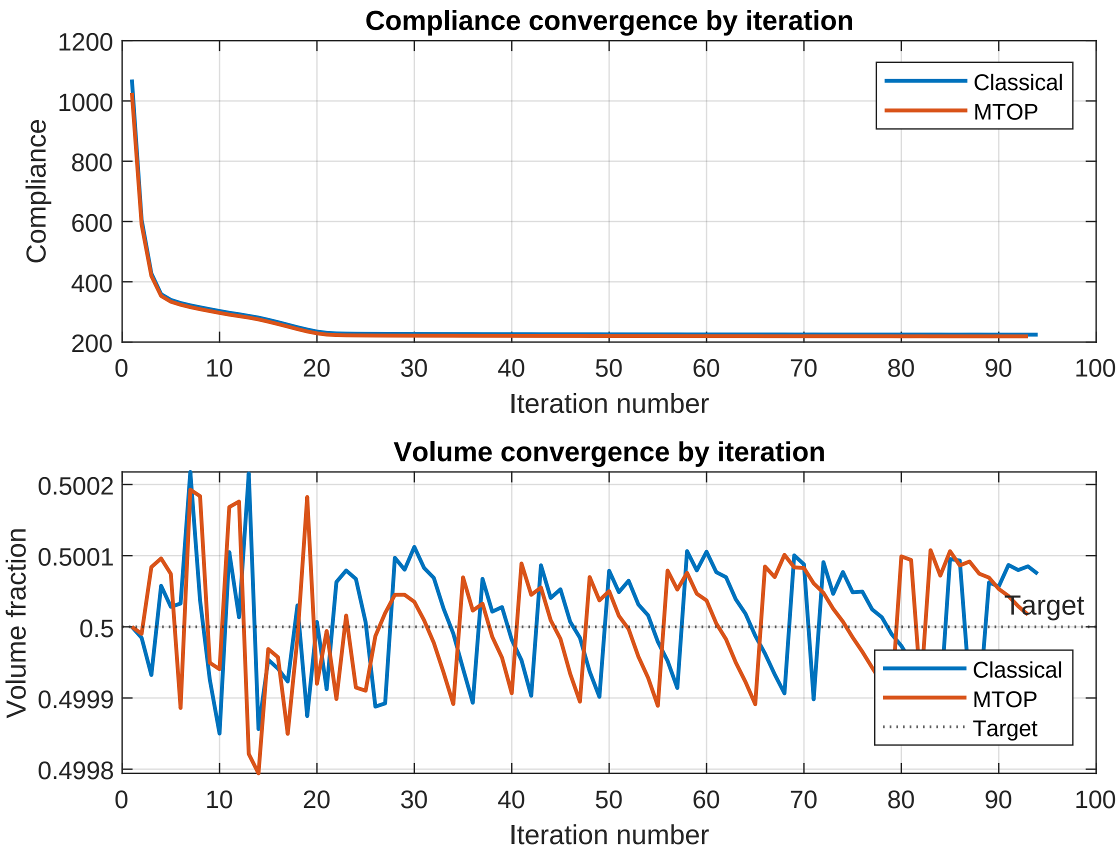

Figure 6: Convergence comparison of the classical and MTOP OC runs as a function of iteration number. Both formulations need almost the same number of OC iterations to satisfy the design-change tolerance.

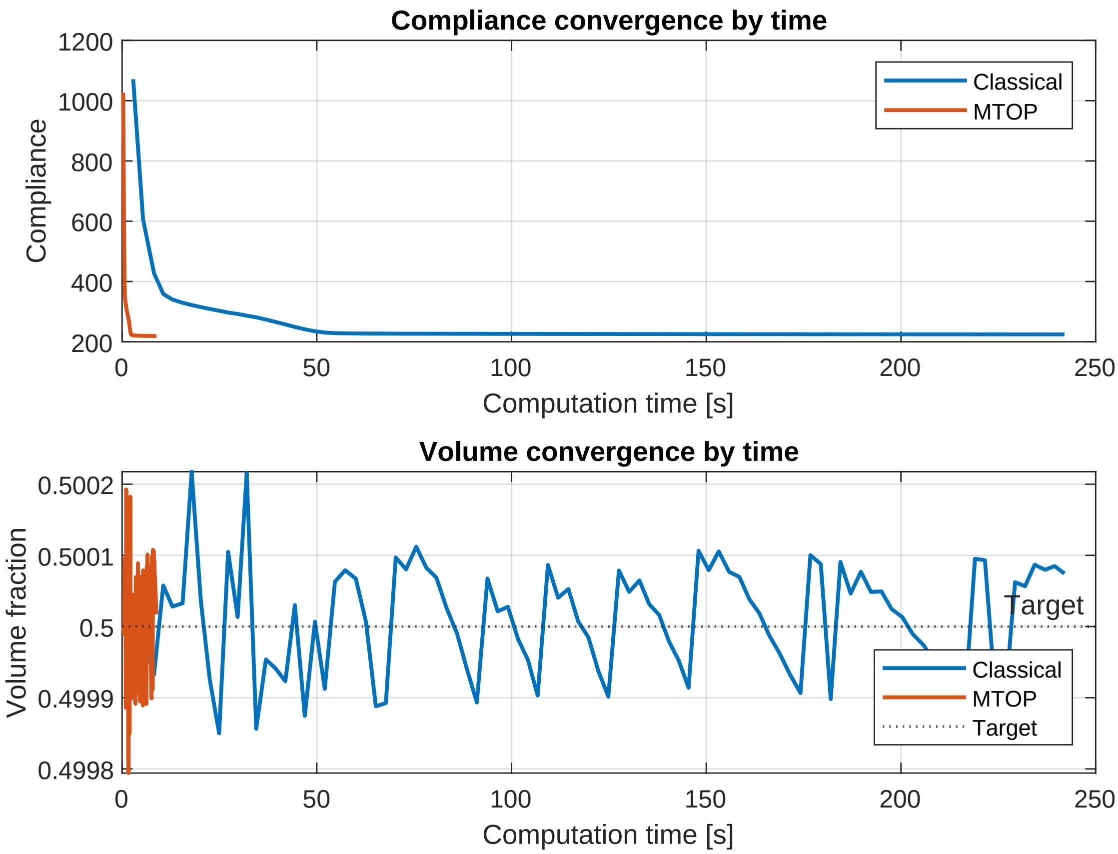

Figure 7: Convergence comparison of the classical and MTOP OC runs as a function of elapsed computation time. The MTOP curve is compressed near the left side of the plot because the equilibrium problem is solved on the 120×40 analysis mesh.

to the prescribed volume fraction, while the compliance history shows a rapid early decrease followed by a slow approach to the final layout.

The MTOP result preserves the same main load paths as the classical solution. The mean absolute density difference is 0.00217, the RMS density difference is 0.00429, and the maximum absolute density difference is 0.03035. This indicates that, for sensitivity filtering with a radius of 24 density cells, the coarse 120×40 analysis mesh does not visibly change the final density layout. The fine-mesh compliance evaluation confirms this visual comparison: the MTOP design is only 0.0357% higher in compliance than the classical design when both are evaluated on the 600×200 finite element mesh.

The iteration counts are almost identical, with 94 iterations for the classical baseline and 93 iterations for the MTOP run. The main difference is computational cost. The MTOP optimization took 8.82 s, compared with 241.99 s for the classical baseline, giving a wall-clock speedup factor of 27.43 for this run. This speedup comes from solving the equilibrium equations on the coarse analysis mesh while keeping the same density resolution for the design description.

The remaining discussion will compare these sensitivity-filtered OC results with the alternative filtering and optimizer choices. Important points to address include:

- how sensitivity filtering and density filtering affect the comparison,
- how MMA and Heaviside projection influence discreteness, convergence, and computational cost.

10 Use of generative AI

Generative AI was used to help structure the repository, draft the report outline, and plan the MATLAB implementation workflow. All numerical formulations, code behavior, sensitivities, results, figures, and interpretations must be checked and validated by the authors. The final report should state exactly which parts were supported by GenAI and how correctness was verified.

11 Conclusion

Assignment step 1 established the classical OC reference solution for the 600×200 MBB beam with sensitivity filtering. Assignment step 2 shows that the MTOP formulation with a 120×40 finite element mesh and 5×5 density cells per finite element reproduces nearly the same density layout while reducing the measured optimization time by a factor of 27.43. The final MTOP design remains essentially equivalent to the classical design when it is evaluated on the fine finite element mesh.

References

[1] T. Nguyen, G. H. Paulino, J. Song, and H. L. Chua. A computational paradigm for multiresolution topology optimization (mtop). *Structural and Multidisciplinary Optimization*, 41(4):525–539, 2010.